THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2008 / CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 12, 2014 9:30 am – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) A crack on a section of the railway track has been discovered during a routine inspection. Since the defect may pose a threat to passengers' safety, the defective section of the railway track must be replaced immediately. Explain the concept of lump sum contract and discuss whether lump sum contract lends itself to emergency repair work. [7 marks]

   b) Propose with detailed reasons another contract type that can effectively deal with the problem in (a) and describe the mechanism of this type of contract. [10 marks]

   c) In order to maintain a reliable and safe service, the railway company has set out a programme to progressively replace the ageing railway tracks. Discuss which contractual arrangement is the most suitable for planned maintenance work. [8 marks]

2. a) A new international airport is going to be constructed in a developing country. Given the scale and complexity of the project, two contractors formed a joint venture company specifically for this project which has subsequently won the tender of this project. Explain to the new joint venture company the key characteristics of functional, project-based and matrix organisation. [18 marks]

   b) Discuss which organisation structure would serve best for this project and prepare a simple organisation chart based on the selected form of organisation setting. [7 marks]

3. a) An investor would like to acquire a land for the construction of a hotel. The land being targeted permits a gross floor area of 5,250m^2 be built despite the hotel rooms would only occupy around 4,000m^2 of the floor area. The hotel will take 24 months to build with the construction cost estimated at \$18,000/m^2. It is anticipated that the income will be around \$225,000 per day in the first three years, with a possible increase in income by 15% subsequently. The operating cost will be \$120,000 per day. A short-term finance at a rate of 10% per annum is needed. The yield for hotel of similar grade and location is 8%, with an expected profit of 20% of development value. Advise the investor the maximum land price they could pay. [20 marks]

   b) Explain to the investor why the discounted cash flow methods are not applicable in computing the maximum land price. [5 marks]

4. a) Highlight five circumstances in which Engineer's Instruction can be issued under the General Conditions of Contract for Civil Engineering Works. [5 marks]

   b) The contractor has submitted a cost estimate for an additional work as specified in the variation orders. The cost estimate relating to the new 150mm thick concrete slab is shown in Appendix I. As the Project Engineer, check the cost estimate and prepare a revised estimate based on your professional knowledge and judgement. A preliminary enquiry reveals that the price of materials and labour wages as shown in the contractor's cost estimate are in line with those in the market. [20 marks]

5. a) A critical task of project planning is to ensure the necessary resources and information are available at the right time during different project stages. Explain why planners should strive to coordinate the resources and information as listed below and prepare the necessary schedules to illustrate how these resources and information can be systematically coordinated:

      i) Coordination of information
      ii) Coordination of materials
      iii) Coordination of various parties involved in the project
      iv) Coordination of plant and equipment [20 marks]

   b) Outline the reasons of delay caused by the client and the contractor respectively and explain the difference in consequence when the delay is induced by these two parties. [5 marks]

~ END OF PAPER ~

Insitu Concrete (1:2:4, 40mm Aggregate), 150mm Thick Slab

Description			$/m^3	$
Material				
Cement @ $800.00/tonne			800.00	
Sand @ $120.00/tonne			120.00	
Aggregate $100.00/tonne			100.00	
			1,020.00	
Add	10% Inflation (due to possible price increase)		102.00	
			1,122.00	1,122.00
Plant			$/m^3	
Mixer, hired (ouput of mixer = 3 m^3/hour) @ $150/hour			150.00	
Crane and skip - $3,000/hour			3,000.00	
		Total plant cost =	3,150.00	3,150.00
Labour			$/m^3	
Mixer driver				
1 number x $100.00/hour			100.00	
Gang		$/hr		
2 x placers @ $100.00/hr		200.00		
2 x levellers @ $100.00/hr		200.00		
1 x vibration @ $120.00/hr		120.00		
		520.00		
Gang output = 1.17 hours / m3				
$520.00/hour ÷ 1.17 hours			444.44	
		Total labour cost =	544.44	544.44
		MATERIAL + PLANT + LABOUR :		4,816.44
		150mm thick slab :		722.47
		Unit Rate:		**$722.47/m^2**